

Fraud Detection Internship Project Summary

- Transaction and identity CSV files were loaded and merged on TransactionID.
- Columns with more than 50% missing values were dropped.
- Remaining numerical values were imputed using median and categorical values using mode.
- Feature engineering included AmtToMeanRatio, HourOfDay, and DeviceRisk.
- LightGBM, XGBoost, and Isolation Forest were trained and compared.
- Evaluation used Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC.
- SHAP was used for global and local explainability.
- Risk tiers were created: Critical Risk, Suspicious, and Clear.
- A Streamlit dashboard was created for operations, exploration, and SHAP explanations.
- Recommended policies include step-up verification and real-time device/hour monitoring.